How can we invest in healthcare as infrastructure?

An infrastructure mental model for investors in antimicrobial products, vaccines, and diagnostics

The infrastructure sector, electricity grids, gas networks, airports, water utilities, solved a structural problem that healthcare still has, decades ago: how do you finance expensive, long-lived assets such as antimicrobials that must be available on demand, where overuse is systemically damaging, and where society cannot tolerate failure in a crisis? The answer was capacity markets, regulated allowed returns, and a financing architecture built around availability rather than consumption. This piece distils a conversation with the head of the Global Listed Infrastructure Van Lanschot Kempen Investment Management, Jags Walia, and suggests a set of practical frameworks for applying exactly that logic to antimicrobial products, and, crucially, to the diagnostic and vaccine platforms without which the whole system cannot function.

This piece also builds on earlier work framing antimicrobial effectiveness as innovation security[1], and on the 2026 Antimicrobial Resistance (AMR) Benchmark findings[2] that the pipeline is thinning while resistance rises. At the same time, a lack of access to and frequent shortages of existing antimicrobials, vaccines and diagnostics continue to prove a major global challenge. Despite significant attention and some notable progress, the case for investing in antimicrobials through a traditional biotech and pharmaceutical investment lens remains weak. The infrastructure framing may be where the investment logic lives; therefore, it must move from a theoretical perspective to a useful investment mental model to support investment research and analysis of companies in the antimicrobials space.

Reframing antimicrobials, diagnostics, and vaccines as infrastructure could unlock a different set of investment tools and expectations. Infrastructure investment is characterised by long-dated debt, inflation-linked returns, and a tolerance for low but stable growth compared to traditional biotech and pharmaceutical investing. This forms a profile well matched to the public health need for products that must remain available, effective, and affordable over the long term. This could help unlock access to new sources of capital to support sustainable development, manufacturing and supply in an area that is currently held back by a significant lack of investment.

A new audience for this argument may not be just the companies producing antimicrobials themselves. It is the companies whose business models depend on antimicrobials continuing to work, and whose investors have not yet priced the exposure. A large proportion of cancer patients require antibiotics to survive chemotherapy, and cancer patients face roughly three times the risk of a drug-resistant infection compared with the general population[3]. As resistance grows, treatment protocols are suspended when infections take hold, and the addressable market for oncology products quietly shrinks. Surgical mortality risk rises with resistance rates, which means that the long-term market for robotic surgery platforms, laparoscopic instruments, and theatre equipment is partially a function of whether antimicrobials remain effective. Organ transplant programmes rely on sustained immunosuppression, which is only clinically viable when post-operative infection risk is manageable. Advanced immunotherapies activate immune pathways that simultaneously raise infection risk. None of these companies describes itself as being in the antimicrobial space. All of them carry an unmodelled exposure to it.

The basic mental model: from pills to capacity

Think about an antimicrobial not as a box of pills but as a fire station. You do not pay firefighters per fire; you pay them to be there, trained, equipped and mostly idle. That idle capacity is the point. Electricity markets already price this logic directly. In the US, power producers receive annual payments simply to commit megawatts of available capacity that can be called on in stress scenarios, regardless of actual generation. Regulators then set an 'allowed return' on that capacity, ensuring a fair but not excessive return on capital.

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[1] The hidden infrastructure of modern medicine: why untreatable infections are one of the next frontiers of system-level investing
[2] The innovation dependency investors are still not pricing
[3] Gupta, V. *et al.* 2025. Incidence and prevalence of antimicrobial resistance in outpatients with cancer: a multicentre, retrospective, cohort study. The Lancet Oncology 26: 620-628. - link

This is the 'Reliability Pricing Model', which ensures long-term grid reliability by procuring the appropriate amount of power supply resources needed to meet predicted energy demand years in the future. Under the 'pay-for-performance' model, companies must deliver on demand during system emergencies or owe a significant payment for non-performance. It is an insurance policy: for a small additional cost (payment to resources which perform well), consumers will have greater protection from power interruptions and price spikes during weather extremes. By matching power supply with future demand, the company's capacity market creates long-term price signals to attract the needed investments to ensure adequate power supply.

The conventional drug model runs in the opposite direction: revenue is a function of price and volume and success comes from higher sales. For antimicrobials, the public health objective is the exact opposite: low volume, targeted use, preserved effectiveness over decades. This structural contradiction, that the only way a manufacturer gets paid is by selling more pills, is what the Wellcome Trust and others have described as a broken market[4]. The capacity model flips this: companies are paid to maintain effective coverage, not to drive unit sales. The UK's antimicrobial products subscription model (a fixed annual NHS payment decoupled from volume) is the closest real-world example. It is in its infancy, but the logic is identical to capacity payments in electricity, and several other countries are following it. Analogously to infrastructure investing, companies adopting the subscription model can also grow through improved efficiency and innovation, offering plenty of growth opportunities beyond simply capacity and service reliability – more on this below.

The UK subscription model is rightly held up as a key example of creating a suitable market for antimicrobials[5]. But there is a problem with treating the UK subscription model as the destination rather than the starting point. To start, the model is not novel at all; it is simply novel for healthcare.

For example, capacity payments are annual fees paid to power generators simply to keep capacity available and dispatchable, regardless of how much electricity they actually produce, and have governed electricity markets for decades. Regulated asset bases, the stock of capital assets on which a utility is permitted to earn a defined, regulator-approved return, independent of short-term demand fluctuations, underpin airport and water utility financing across Europe. Another one is take-or-pay contracts, which are agreements in which a buyer commits to pay for a minimum volume whether or not they draw it, transferring demand risk from the supplier to the offtaker, and are the standard commercial structure for gas pipelines. In each case, the logic is identical: separating the right to earn a return from the act of selling units, because the asset's value to society lies in its availability, not its throughput.

The AMR space treats these mechanisms as aspirational endpoints, whereas a more productive posture is to treat them as established templates. The detailed architecture already exists, and the task is continued translation, not just to the level of policy, but for companies and potentially also to investors. Noting that investors who already hold regulated infrastructure understand the risk/return profile of capacity payments intuitively, and the case to be made for generalist investors or healthcare specialists is that antimicrobial effectiveness is *also* part of the same infrastructure asset class, but expressed through biology and public health needs, rather than steel, wire, and transportation or energy needs.

What needs to come together to make investing in antimicrobial companies viable is the right market mechanism, product strategy and investor outlook. Together, this can build a viable business case for antimicrobial innovation and supply. Alongside the need for more stable market systems, what remains underdeveloped is the work of showing how an antimicrobial company with the right contract structure, supply resilience, and stewardship approach can be re-evaluated at a lower discount rate; how its cash flow profile maps onto the duration preferences of pension funds and sovereign wealth funds; and how the blended finance structures already used in infrastructure, such as Development Finance Institution (DFI) first-loss tranches, sovereign guarantees, Advanced Market Commitments (AMCs), can be imported into this space. That translation is the frontier.

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[4] Reframing Resistance | Reports | Wellcome

[5] Outterson, K.; Rex, J. H. 2023. Global Pull Incentives for Better Antibacterials: The UK Leads the Way. Applied Health Economics and Health Policy 21 (3): 361-364. - link

It is worth emphasising that the infrastructure framing is powerful, but it has important limits that investors should understand. The development, regulatory approval, and commercialisation of a new antimicrobial, diagnostic, or vaccine entail scientific and clinical risks that have no direct equivalent in traditional infrastructure investing. Assets can fail in trials, regulatory bodies may require additional data, and approved products can face post-market safety findings that alter their commercial trajectory. The implication is that the infrastructure framing applies most cleanly to existing approved products or to late-stage clinical assets where the probability of technical failure has fallen substantially, and capital requirements have risen sharply. This is also precisely the stage at which infrastructure-style financing becomes most appropriate, as the cash flow profile becomes more predictable and the risk profile begins to resemble a project finance opportunity rather than a venture bet. Earlier-stage development is better suited to the public funding, philanthropic capital, push incentives and venture capital funding that already exist to carry pre-commercial risk, with infrastructure capital stepping in as products approach and clear regulatory approval.

Finally, traditional infrastructure investing is not without its own regulatory risk. Periodic price control resets can reduce allowed returns, policy shifts can alter permitted revenue models, and, particularly in energy, stranded asset risk is a well-modelled exposure. Infrastructure investors are therefore not naïve to regulatory uncertainty; they price and manage it. The difference is that in conventional infrastructure, the asset itself is not at risk of scientific failure. In antimicrobials, both dimensions of risk are present simultaneously, which argues for structuring investment vehicles that explicitly separate the two: public and blended finance absorbing scientific risk in early development, and private infrastructure capital entering once that risk has been substantially retired.

What makes an infrastructure-grade AMR company?

If antimicrobials are infrastructure, then some developers and manufacturers can acquire an infrastructure-like character on top of their healthcare one, with consequences for valuation, discount rates, and portfolio fit. The following is a practical checklist drawn from how infrastructure funds evaluate assets.

One clarification matters before proceeding: the framework applies equally to producers of antimicrobial therapeutics, vaccines that reduce infection incidence, and diagnostics that enable targeted and appropriate prescribing. The infrastructure is not just the molecule; it is the system. A rapid diagnostic test that prevents five courses of unnecessary broad-spectrum antibiotics from being prescribed is as much a part of antimicrobial effectiveness infrastructure as the antibiotic itself; a vaccine that reduces the incidence of bacterial pneumonia reduces pressure on the entire antimicrobial system to resist. This means the investment checklist below is relevant to companies across all three modalities, and the green and red flags that follow apply equally to all.

1. A defensible position in a defined pond

Infrastructure is often locally monopolistic. Heathrow has global competition, but it is the only major airport in West London. The AMR analogue would not be a global monopoly on a molecule, but a regulated local/regional position backed by a long-term contracted positions in defined geographies with barriers to rapid substitution that make a company the effective 'Heathrow of antimicrobials X' in country Y. To identify this, an investor question could be: *How concentrated is the company's position, and how much of forward revenue is contracted rather than speculative?*

2. Resilient capacity: the ability to turn it on when needed

Utilities brag about resilience: two runways, gas generation on standby, *etc.*. The COVID-19 pandemic made the equivalent argument for medicine with uncomfortable force. A significantly large volume and variety of Active Pharmaceutical Ingredients (APIs) used in high-income markets are manufactured in a small number of facilities concentrated in China and India[6], a supply geography that was not seriously interrogated until borders closed, freight costs spiked, and hospitals began rationing drugs they had assumed were permanent fixtures. Antimicrobials were among the most exposed categories: generic antibiotics have thin margins,

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[6] Global manufacturing capacity for active pharmaceutical ingredients remains concentrated

long lead times, and limited buffer stock, which meant that even a modest demand surge or logistics disruption could rapidly translate into clinical shortages.

The EU's current response through the Critical Medicines Act[7], mandating that a defined share of critical medicines be sourced from EU producers and classifying antimicrobial supply as a strategic national interest is a policy acknowledgement that supply geography is a systemic risk, not merely a procurement preference. For AMR companies, resilience therefore means more than having a backup facility on a list. It means tested surge plans with documented time-to-double output; dual-sourced or regionally diversified inputs for key APIs and excipients; fallback fill-finish capacity that does not depend on a single contract manufacturer; and buffer stock strategies calibrated to realistic shock scenarios, not peacetime demand. These resilience strategies will vary by product type, particularly between high-volume generics and low-volume branded products, and need to be thought through carefully as part of a company's business case to support engagement with infrastructure investors.

A company that can credibly answer the question "*what is your maximum sustainable stress run, and what is the first thing that fails?*" is structurally different from one that cannot and should be valued differently. This is precisely the kind of operational disclosure that infrastructure investors routinely demand and that investors may have rarely considered asking for from healthcare companies. To identify this, an investor question could be: *"If a regional outbreak hit tomorrow, how fast can this company ramp, and what breaks first?"*

3. Ownership of critical assets

Airports and pipelines own the fixed assets that deliver their services (e.g., parking lots, restaurants and shop areas, *etc.*). The AMR equivalent is ownership or long-term control of API manufacturing, fill-finish capacity, and quality/sterility facilities through leases or take-or-pay contracts. Whilst this is not always possible, particularly for smaller biotech companies, when production is outsourced, the contracts could be set up to resemble infrastructure offtakes rather than spot sourcing. Therefore, demonstrating strong contractual durability and control over necessary assets. To identify this, an investor question could be: *If you map what is truly owned versus what is contracted. Where are the single-source dependencies?*

4. Long-dated, volume-independent cashflows

Regulated grid operators may have 30-50 years of predictable allowed returns on their regulated asset base. The AMR equivalent would be 10- to 15-year subscription contracts with health systems, indexed payments (inflation-linked), and multi-country or supranational agreements spreading political risk. The valuation implication is direct: a lower, more stable perceived risk of future cash flows justifies a lower discount rate and a higher multiple. This is a different route to value creation: not 'more volume, more price', but the same growth, drastically de-risked. To identify this, an investor question could be: *What proportion of forward revenue (5-10 years) is contracted versus speculative?*

How infrastructure companies grow, and why that is appealing to long-term capital

A reasonable concern is often expressed along the lines: *"If you are incentivised to sell less, how does this company grow?"* The answer is that it grows like regulated infrastructure, not like a volume-driven pharma business. The growth levers are:

- Geographic expansion: winning capacity subscriptions in more countries for the same asset base, like a grid company extending its regulated asset base into new regions.
- Scope expansion: moving from capacity for a single drug in one indication (e.g., sepsis) to another use (e.g., surgical prophylaxis, resistant niches for specific diseases, and other priority indications) and/or to a broader portfolio of assured effective coverage, spanning therapeutic antimicrobials, diagnostic platforms that enable appropriate prescribing, and vaccines that reduce the infections requiring treatment in the first place. Each of these reduces resistance pressure on the system as a whole, which in turn extends the economic life of contracted capacity. A company that co-develops the diagnostic alongside the antibiotic, or that partners with a vaccine manufacturer to protect the

[7] Critical medicines: EU measures to boost competitiveness and tackle shortages

efficacy of its portfolio, is pursuing the same logic as a grid operator that invests in demand management to extend the life of its generation assets.

- Efficiency gains: in underprice-cap style models, part of the efficiency gains from process innovation (i.e., continuous manufacturing, greener chemistry, recycling of unused compounds) accrue to shareholders as excess returns until the next regulatory reset, exactly as in UK utilities[8] and some European airports[9].
- Adjacent services: data and surveillance infrastructure, AMR analytics, creation of stewardship tools. These become high-margin service lines, bundled with drug capacity contracts.
- Infrastructure-style funding (debt-financed capacity at lower cost of capital): infrastructure assets are routinely funded through long-dated project finance debt rather than equity alone, precisely because stable, contracted cash flows support investment-grade credit ratings and reduce the Weighted Average Cost of Capital (WACC). A gas pipeline backed by a 20-year take-or-pay contract can raise debt at spreads unavailable to a speculative pharmaceutical developer because the revenue risk profile is fundamentally different. The same logic applies to an antimicrobial company that has converted a meaningful share of its revenue into multi-year subscription contracts with national health systems: the contracted cash flows should support a higher debt capacity, a lower cost of capital, and therefore a higher equity value at the same level of operating earnings. This is not hypothetical; it is the standard infrastructure financing playbook, and it has been applied in blended finance structures in which DFI or sovereign guarantees provide first-loss protection to attract private debt to otherwise 'unbankable' assets. The AMR Action Fund and GARDP's (the Global Antibiotic Research & Development Partnership) co-development models are early-stage versions of this logic, funding and performing product development with lower than industry-standard return expectations. The next step may be to extend it into the capital structure of the operating companies themselves.
- The infrastructure analogy also extends to the treatment of innovation expenditure. In regulated utilities, efficiency innovations such as smart grid technology, advanced metering, and digital fault detection are treated not as speculative research and development (R&D) bets but as capital investments in an existing asset base, recoverable through the regulated asset base (RAB) with an allowed return. The equivalent for antimicrobial infrastructure would be investment in next-generation modalities, such as CRISPR-based bacteriophage therapies, monoclonal antibody treatments targeting resistant pathogens, and rapid point-of-care diagnostic platforms. Under a capacity framework, these would not compete with existing molecules on a volume-based return model; they would expand the scope of 'assured effective coverage' that the company can contract with health systems, analogous to a grid operator adding renewable generation capacity to its regulated asset base. Companies already moving in this direction are building what will eventually look, from a financing perspective, like modular infrastructure upgrades rather than high-risk pipeline bets.

In most cases, all these types of growth would compound at modest (some possibly higher) but decidedly predictable rates, precisely the risk/return profile that appeals to long-dated liabilities: pension funds, sovereign wealth funds, and insurance balance sheets. As governments and health systems consider how to build effective markets for AMR products, they should consider how best to align with these growth principles.

Green flags and red flags

At the end of the conversation with Jags, I focused on listing a practical filter for separating infrastructure-grade AMR companies from those trapped in an undervalued model. See the green and red flags below.

Green flags

i. Long-dated, volume-decoupled contracts with health systems or supranational buyers.
ii. Local or regional manufacturing for critical drugs, with tested surge capacity and documented fallback facilities.

[8] Energy price cap explained | Ofgem
[9] Aviation Act governing the operation of Schiphol

iii. A board-level narrative framing the company as a ‘health security’ or ‘infrastructure’ provider, not purely as an antimicrobial products seller.
iv. Transparent reporting: resistance trends for own molecules, stock-out rates and shortage monitoring, plant reliability, water and sanitation and infection-control partnerships.
v. Co-investment in different pillars, for example, a therapeutics producing company investing in vaccines and/or diagnostics, and infection prevention projects in key markets, explicitly to protect the effectiveness of their portfolio (this is asset-protection capex, not philanthropy).

Red flags

i. Strategy documents framed entirely around volume growth and market share in antimicrobial products.
ii. Aggressive sales targets in regions with weak stewardship and no parallel investment in diagnostics or clinician education.
iii. Single-region or single-plant dependence for critical APIs, justified solely by short-term efficiency.
iv. Frequent stock-outs in lower-income markets, with no credible remediation plan.
v. R&D portfolio dominated by derivative molecules chasing tender-driven sales, with no effort to embed drugs into broader health-system resilience.

Making it actionable

For investors

- When analysing antimicrobial-producing companies, evaluate them through an infrastructure lens. Explicitly adjust discount rates where a company has on-shored resilient capacity, long-term volume-independent contracts, and a credible stewardship strategy.
- Ask for infrastructure-style disclosures: surge capacity metrics; resistance prevalence for own molecules in key markets; share of revenue from stable, multi-year contracts versus spot tenders.
- Engage on contract structure. Encourage management to pursue subscription/capacity models, such as the UK subscription model, and to proactively propose them to payers.
- Support blended-finance structures. Back capital allocation decisions from companies that combine public guarantees with private capital for on-shored AMR infrastructure, particularly in lower-income markets, where public and private partnership-style structures work well.
- Engage adjacent sector holdings (e.g., oncology companies, surgical robotics businesses, organ transplant specialists) on quantified dependency. Companies that understand this dependency earliest will build it into their growth strategy and investor narrative, and will be better positioned to address the impact of the rise of untreatable infections on their business models.

For companies

- Expand the corporate identity: stop describing the business as ‘antimicrobial manufacturer’ alone and articulate an added strategy around ‘health security infrastructure’.
- Lobby smarter. Do not ask governments simply for more incentives. Ask for regulatory frameworks that define allowed returns on AMR capacity, with clear performance KPIs, importing what already works in utilities and airports, for example. Discuss infrastructure-style funding possibilities, too.
- Invest in stewardship as asset protection. Co-fund diagnostic rollouts and vaccination campaigns in key markets. These actions are part of a renewed message, something along the lines of “*We don’t just sell the molecule; we sell reduced resistance risk.*”

So, in a nutshell, the AMR space often behaves as if it must invent an entirely new economic model from scratch. But it may be that the homework has already been done by electricity grids, gas networks, airports, and water utilities. Their answer is capacity markets and regulated allowed returns. So, the opportunity is to import that investment framework into AMR, with appropriate tweaks for biology and global equity. Antimicrobials are not just drugs. We should pay for them the way we already pay for infrastructure: for capacity, resilience and readiness.

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